



Verifying ARP

Difficulty (out of 5 stars): ★★★

Recommended study time: 2 hours

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1 About This Lab

1.1 Task Description

On a LAN, a host or a Layer 3 network device must learn the network layer address (that is, IP address) of the destination host or Layer 3 network device before sending data to it. However, the IP address alone is not enough. IP packets must be encapsulated into frames before being sent over a physical network. The sender must identify the receiver's physical address (MAC address), requiring a mapping between the IP and MAC addresses. The Address Resolution Protocol (ARP) can resolve IP addresses into MAC addresses.

After completing this task, you will be able to clearly understand ARP and independently use ARP for network verification.

1.2 Objectives

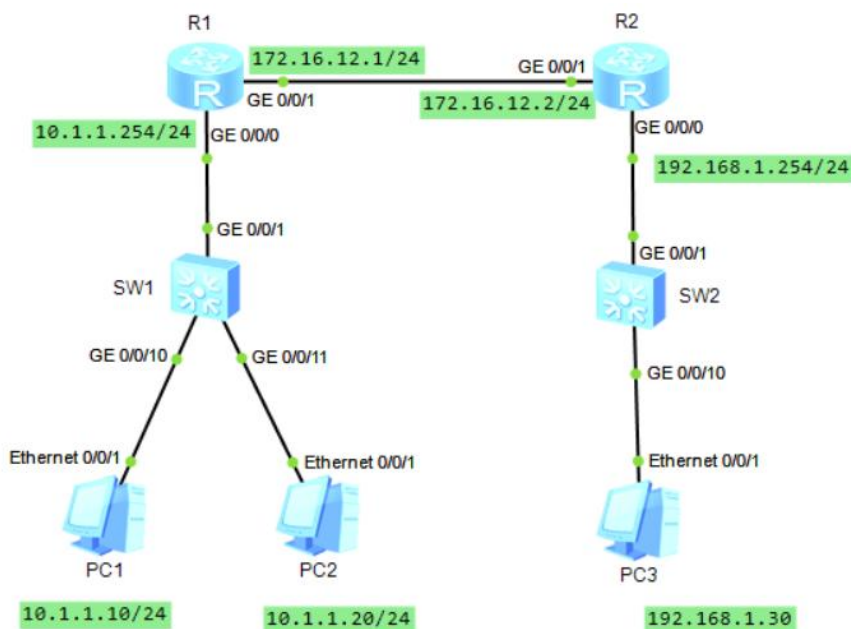
On completing this task, you will be able to:

1. Understand ARP fundamentals.
2. Understand how hosts on the same network communicate with each other using ARP.
3. Understand how hosts on different networks communicate with each other using ARP.

2 Task Preparation

2.1 Network Topology

Figure 2-1 Lab topology





2.2 Initial Configuration

1. IP addresses

- Initial configuration of R1

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R1
[R1]int g0/0/0
[R1-GigabitEthernet0/0/0]ip address 10.1.1.254 24
[R1-GigabitEthernet0/0/0]quit
[R1]int g0/0/1
[R1-GigabitEthernet0/0/1]ip address 172.16.12.1 24
[R1-GigabitEthernet0/0/1]quit
[R1]
```

- Initial configuration of R2

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R2
[R2]int g0/0/1
[R2-GigabitEthernet0/0/1]ip address 172.16.12.2 24
[R2-GigabitEthernet0/0/1]quit
[R2]int g0/0/0
[R2-GigabitEthernet0/0/0]ip address 192.168.1.254 24
[R2-GigabitEthernet0/0/0]quit
[R2]
```

- The initial configurations of SW1 and SW2 are not required. By default, all their interfaces are in VLAN 1.
- Static IP address of PC1: 10.1.1.10/24; gateway address: 10.1.1.254
- Static IP address of PC2: 10.1.1.20/24; gateway address: 10.1.1.254
- Static IP address of PC3: 192.168.1.30/24; gateway address: 192.168.1.254

2. Route configuration

- R1 route configuration

```
[R1]ip route-static 192.168.1.0 255.255.255.0 172.16.12.2
```

- R2 route configuration

```
[R2]ip route-static 10.1.1.0 255.255.255.0 172.16.12.1
```

3 Task Implementation

3.1 ARP Application on the Same LAN

PC1 and PC2 are on the same LAN. Ping PC2's IP address from PC1.

Step 1 Run the **arp -d** command to clear the ARP caches of PC1 and PC2, and run the **arp -a** command to check the ARP caches of PC1 and PC2. Ensure that the ARP caches of PC1 and PC2 are empty. See Figure 3-1 and Figure 3-2.



Figure 3-1 Clearing the ARP cache of PC1

The screenshot shows a window titled "PC1" with tabs for "基础配置", "命令行", "组播", "UDP发包工具", and "串口". The "命令行" tab is active, displaying the following commands and their outputs:

```
PC>ipconfig

Link local IPv6 address.....: fe80::5689:98ff:fe2a:e0c
IPv6 address.....: :: / 128
IPv6 gateway.....: ::
IPv4 address.....: 10.1.1.10
Subnet mask.....: 255.255.255.0
Gateway.....: 10.1.1.254
Physical address.....: 54-89-98-2A-0E-0C
DNS server.....:

PC>arp -d

PC>arp -a

Internet Address      Physical Address      Type

PC>
PC>
PC>
```

Figure 3-2 Clearing the ARP cache of PC2

The screenshot shows a window titled "PC2" with tabs for "基础配置", "命令行", "组播", "UDP发包工具", and "串口". The "命令行" tab is active, displaying the following commands and their outputs:

```
PC>ipconfig

Link local IPv6 address.....: fe80::5689:98ff:fe11:3b41
IPv6 address.....: :: / 128
IPv6 gateway.....: ::
IPv4 address.....: 192.168.1.30
Subnet mask.....: 255.255.255.0
Gateway.....: 192.168.1.254
Physical address.....: 54-89-98-11-3B-41
DNS server.....:

PC>arp -d

PC>arp -a

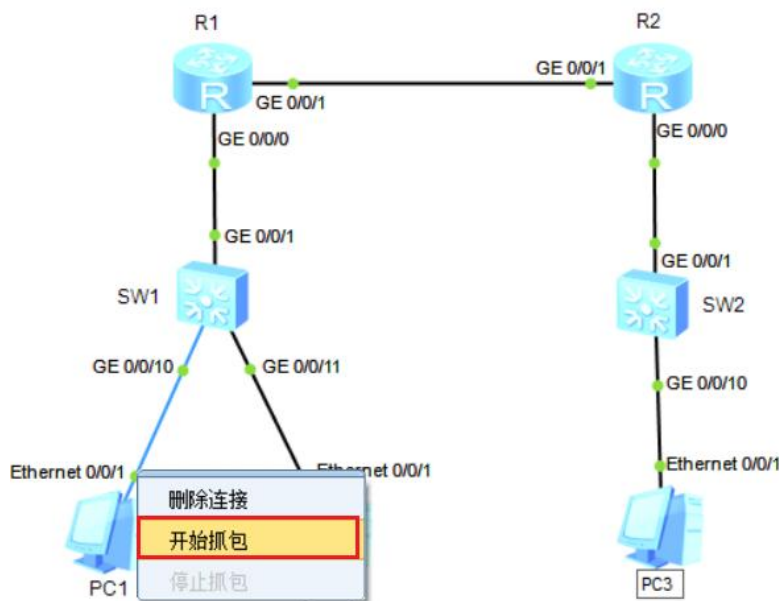
Internet Address      Physical Address      Type

PC>
PC>
PC>
PC>
```



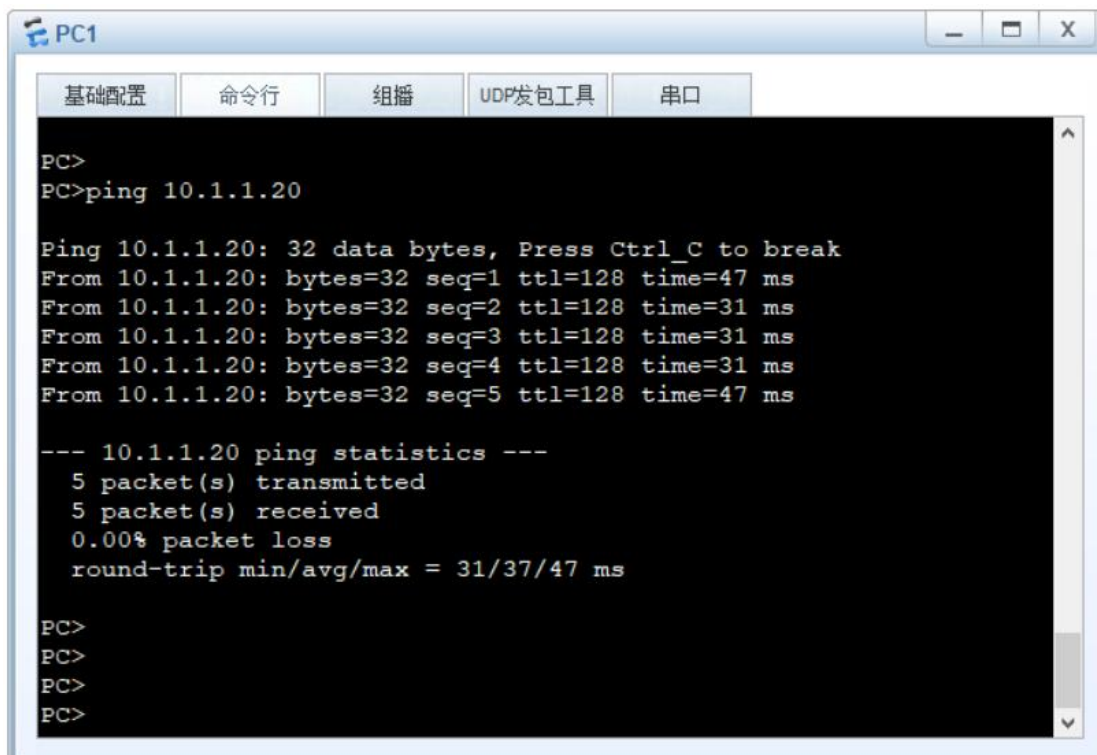
Step 2 Start Wireshark to capture packets on the link between PC1 and SW1, as shown in Figure 3-3.

Figure 3-3 Starting Wireshark to capture packets



Step 3 Ping PC2's IP address 10.1.1.20 from PC1, as shown in Figure 3-4.

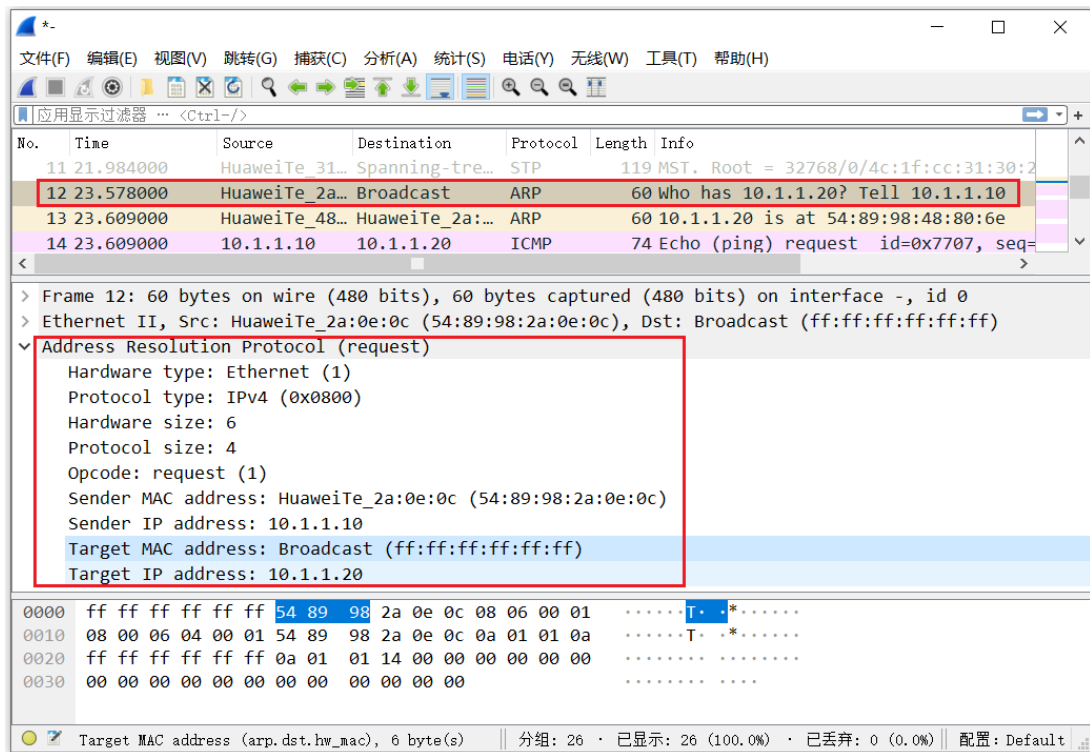
Figure 3-4 Pinging PC2 from PC1





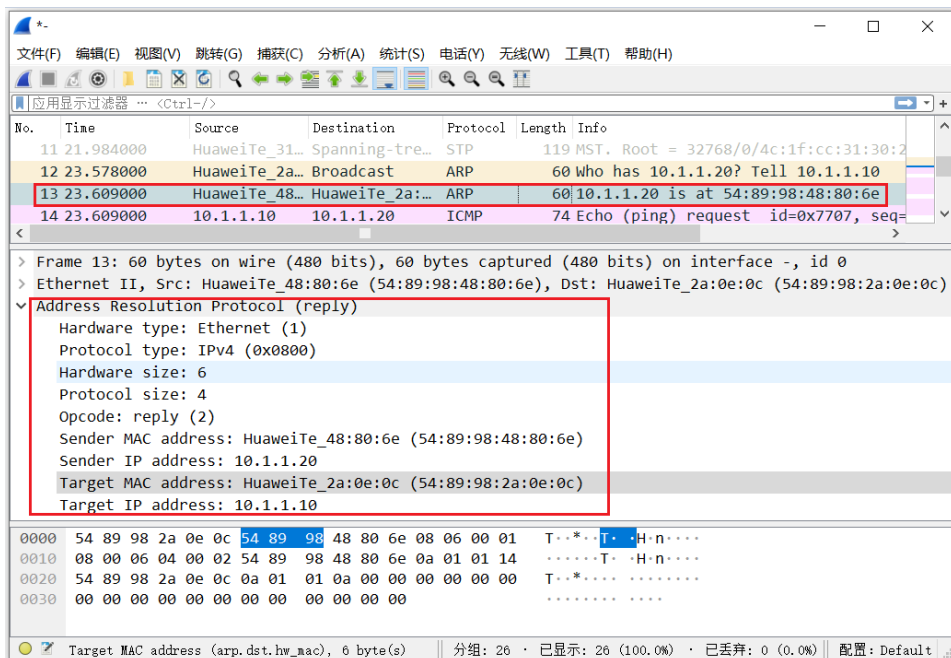
Step 4 View the ARP Request packets captured by Wireshark, as shown in Figure 3-5.

Figure 3-5 ARP Request packets



Step 5 View the ARP Reply packets captured by Wireshark, as shown in Figure 3-6.

Figure 3-6 ARP Reply packets





Step 6 View the ARP cache on PC1, as shown in Figure 3-7.

Figure 3-7 ARP cache of PC1

```
PC1
基础配置 命令行 组播 UDP发包工具 串口
PC>ping 10.1.1.20

Ping 10.1.1.20: 32 data bytes, Press Ctrl_C to break
From 10.1.1.20: bytes=32 seq=1 ttl=128 time=31 ms
From 10.1.1.20: bytes=32 seq=2 ttl=128 time=31 ms
From 10.1.1.20: bytes=32 seq=3 ttl=128 time=47 ms
From 10.1.1.20: bytes=32 seq=4 ttl=128 time=31 ms
From 10.1.1.20: bytes=32 seq=5 ttl=128 time=47 ms

--- 10.1.1.20 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 31/37/47 ms

PC>arp -a

Internet Address      Physical Address      Type
10.1.1.20             54-89-98-48-80-6E     dynamic

PC>
```

----End

3.2 ARP Application Across LANs

PC1 and PC2 are on the same LAN, but PC3 is on a different LAN. Ping PC3's IP address from PC1.

- Step 1** Run the **arp -d** command to clear the ARP cache of PC1, and run the **arp -a** command to check the ARP cache of PC1. Ensure that the ARP cache of PC1 is empty.
- Step 2** Start Wireshark to capture packets on the link between PC1 and SW1.
- Step 3** Ping PC3's IP address 192.168.1.30 from PC1, as shown in Figure 3-8.



Figure 3-8 Pinging PC3 from PC1

```
PC1
基础配置 命令行 组播 UDP发包工具 串口
PC>
PC>ping 192.168.1.30

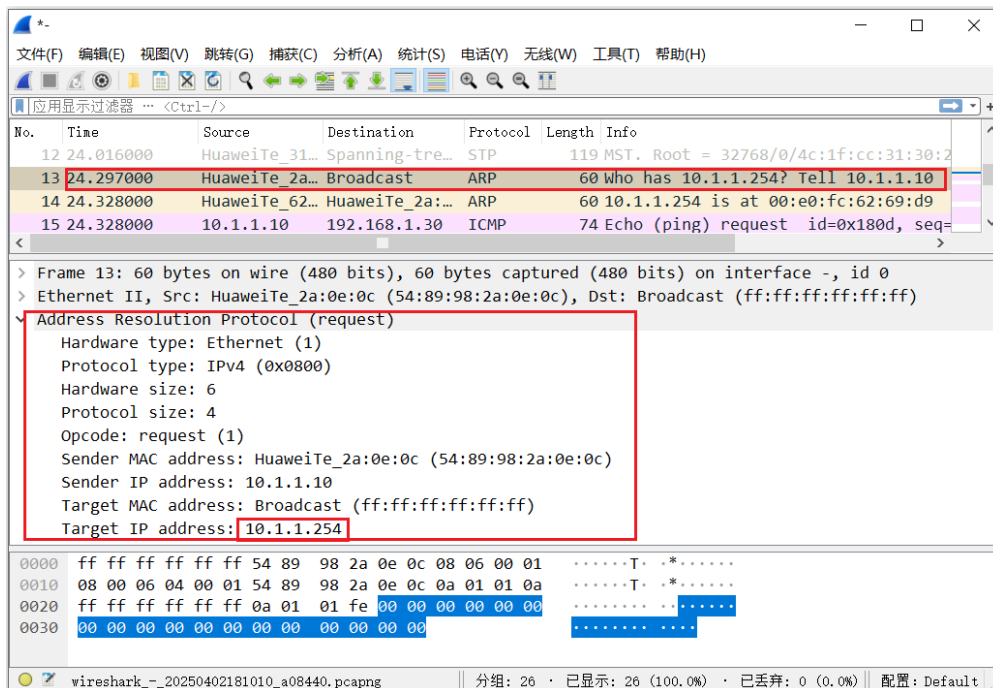
Ping 192.168.1.30: 32 data bytes, Press Ctrl_C to break
Request timeout!
From 192.168.1.30: bytes=32 seq=2 ttl=126 time=47 ms
From 192.168.1.30: bytes=32 seq=3 ttl=126 time=63 ms
From 192.168.1.30: bytes=32 seq=4 ttl=126 time=78 ms
From 192.168.1.30: bytes=32 seq=5 ttl=126 time=78 ms

--- 192.168.1.30 ping statistics ---
 5 packet(s) transmitted
 4 packet(s) received
20.00% packet loss
round-trip min/avg/max = 0/66/78 ms

PC>
PC>
PC>
PC>
PC>
```

Step 4 View the ARP Request packets captured by Wireshark, as shown in Figure 3-9.

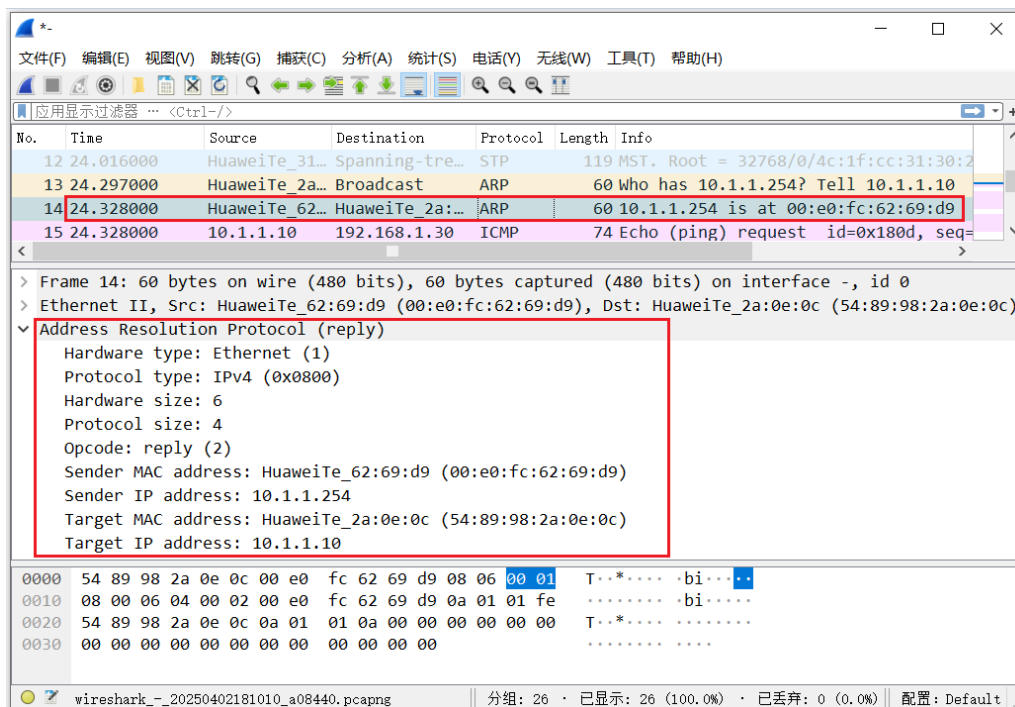
Figure 3-9 ARP Request packets on different network segments





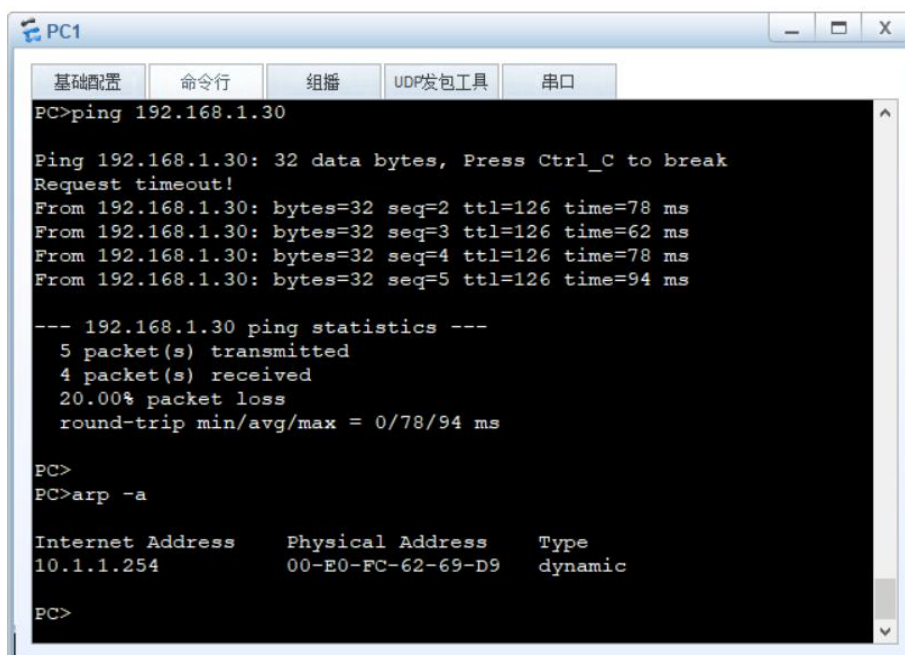
Step 5 View the ARP Reply packets captured by Wireshark, as shown in Figure 3-10.

Figure 3-10 ARP Reply packets on different network segments



Step 6 View the ARP cache of PC1, as shown in Figure 3-11.

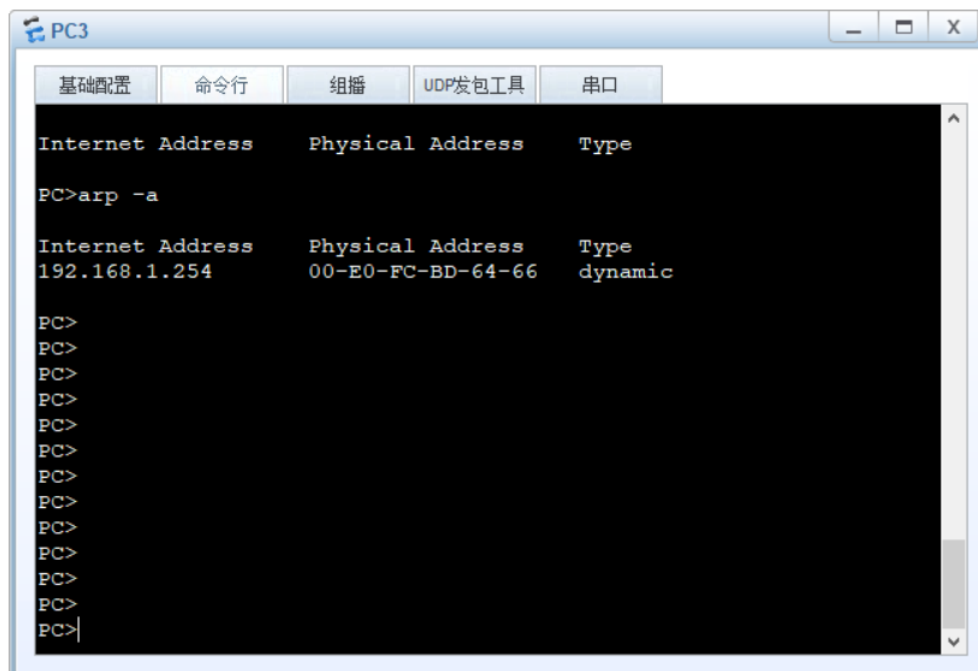
Figure 3-11 ARP cache of PC1





Step 7 Step (7): View the ARP cache of PC3, as shown in Figure 3-12.

Figure 3-12 ARP cache of PC3



----End

4 Verification

None

5 Quiz

1. Describe the ARP communication process between hosts on the same network.
2. Describe the ARP communication process between hosts on different networks.
3. Why is an ARP cache automatically cleared after a period of time during the lab?